SMART CLASSROOM TIMETABLE ALLOCATION

Python Project Report

1.Overview

The Smart Classroom Timetable Allocation system is designed to allocate classrooms to different classes and subjects automatically. It considers classroom availability, class schedules, teacher availability, and classroom capacity while preparing the timetable.

The main purpose of this system is to avoid timetable clashes, reduce manual work, and utilize classroom resources efficiently. It provides a systematic way to assign classrooms to classes at specific times. This system helps educational institutions manage classroom schedules accurately and efficiently.

2.Example Policy Used

The following sample policies are used for classroom allocation:

1. Each classroom can be assigned to only one class in a given time slot.
2. The number of students must not exceed the classroom capacity.
3. Classrooms must be allocated only during available time slots.
4. A classroom cannot be assigned to two different classes simultaneously.
5. The system must check classroom availability before making an allocation.
6. If no suitable classroom is available, the system displays "No Classroom Available."
7. The timetable must display the class name, classroom number, subject, and time slot.
8. The allocation must follow the predefined scheduling rules.

3.Problem Statement

Preparing classroom timetables manually is a time-consuming process. Multiple classes may require the same classroom at the same time, resulting in scheduling conflicts. Limited classroom availability and differences in classroom capacity make timetable preparation more difficult.

Manual scheduling may also lead to errors, inefficient classroom utilization, and unnecessary confusion. Therefore, a smart system is needed to allocate classrooms automatically based on availability and scheduling constraints.

4.Objectives

1. To automate classroom timetable allocation.
2. To avoid classroom scheduling conflicts.
3. To allocate classrooms based on availability.
4. To consider classroom capacity while assigning rooms.
5. To reduce manual work and scheduling errors.
6. To utilize available classrooms efficiently.
7. To generate an organized classroom timetable.
8. To ensure that a classroom is not assigned to multiple classes at the same time.

5.Assumptions

1. The list of classrooms is available in the system.
2. Each classroom has a predefined seating capacity.
3. Class schedules and time slots are known in advance.
4. Each class has a specific number of students.
5. Classroom availability information is accurate.
6. A classroom can accommodate only one class during a particular time slot.
7. The classroom capacity must be greater than or equal to the number of students.
8. All timetable data entered into the system is valid.

6.Algorithm

Step 1: Start.

Step 2: Enter the class details, student strength, subject, and time slot.

Step 3: Enter the available classroom details and their capacities.

Step 4: Select a class that requires a classroom.

Step 5: Check each classroom for availability during the requested time slot.

Step 6: Verify that the classroom capacity is sufficient for the number of students.

Step 7: If a suitable classroom is available, allocate it to the class.

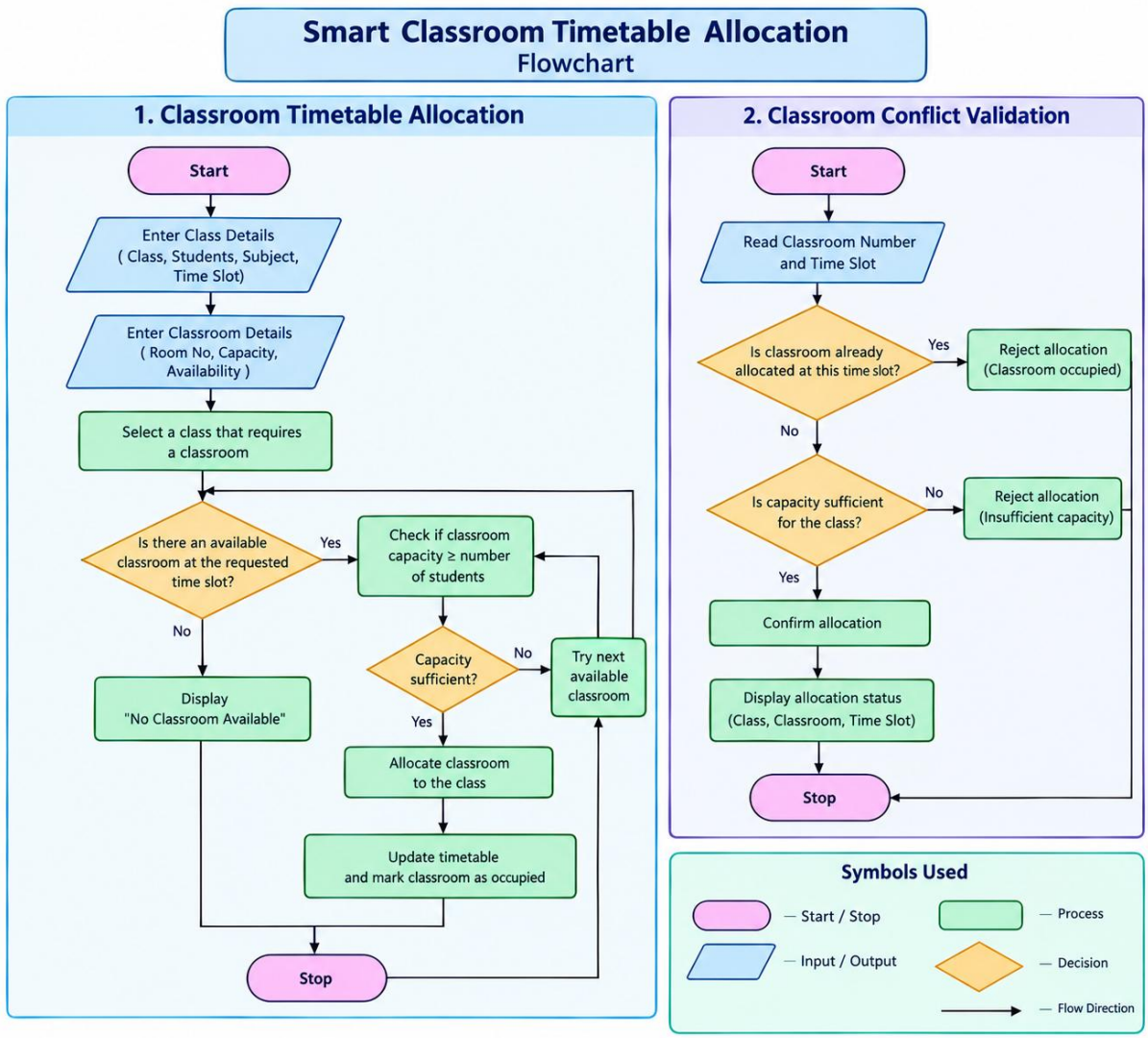
Step 8: Otherwise, search for another available classroom.

Step 9: If no suitable classroom is available, display "No Classroom Available."

Step 10: Store and display the timetable.

Step 11: Stop.

7.Flowchart



8.Pseudocode

START

INPUT Class, Students, Time_Slot

INPUT Classroom, Capacity, Availability

IF Availability = TRUE

  IF Capacity >= Students

    IF Classroom is NOT occupied

```
        ALLOCATE Classroom to Class
        PRINT "Classroom Allocated Successfully"
      ELSE
        PRINT "Classroom Already Occupied"
      ENDIF
    ELSE
      PRINT "Insufficient Classroom Capacity"
    ENDIF
  ELSE
    PRINT "No Classroom Available"
  ENDIF

  STOP
```

9.Source Code

```
# Smart Classroom Timetable Allocation

classrooms = {
    "Room101": 50,
    "Room102": 40,
    "Room103": 30
}

allocations = []

print("SMART CLASSROOM TIMETABLE ALLOCATION")
print("-----------------------------------")

n = int(input("Enter number of classes: "))
```

```
for i in range(n):
    print("\nClass", i + 1)

    class_name = input("Enter class name: ")
    students = int(input("Enter number of students: "))
    subject = input("Enter subject: ")
    time_slot = input("Enter time slot (e.g., 9-10 AM): ")

    allocated = False

    for room, capacity in classrooms.items():

        if capacity >= students:

            occupied = False

            for allocation in allocations:
                if (allocation["room"] == room and
                        allocation["time"] == time_slot):
                    occupied = True
                    break

            if not occupied:
                allocations.append({
                    "class": class_name,
                    "subject": subject,
                    "students": students,
```

```
                "room": room,
                "time": time_slot
            })

            print("\nClassroom Allocated Successfully!")
            print("Class:", class_name)
            print("Subject:", subject)
            print("Classroom:", room)
            print("Time Slot:", time_slot)
            print("Capacity:", capacity)

            allocated = True
            break

    if not allocated:
        print("\nNo Suitable Classroom Available!")
        print("Class:", class_name)
        print("Time Slot:", time_slot)

print("\n========== FINAL TIMETABLE ==========")

if len(allocations) == 0:
    print("No classrooms were allocated.")

else:
    print("{:<12}{:<12}{:<12}{:<15}".format(
        "Class", "Subject", "Classroom", "Time Slot"))
```

```
    print("-" * 55)

    for item in allocations:
        print("{:<12}{:<12}{:<12}{:<15}".format(
            item["class"],
            item["subject"],
            item["room"],
            item["time"]
        ))

print("\nProgram Completed.")
```

10.Test Cases

Test Case ID	Class Name	Students	Time Slot	Expected Output	Status
TC01	CSE-A	45	9-10 AM	Room101 Allocated	Pass
TC02	CSE-B	35	9-10 AM	Room102 Allocated	Pass
TC03	ECE-A	25	9-10 AM	Room103 Allocated	Pass
TC04	CSE-C	45	9-10 AM	Room101 Already Occupied; allocate another suitable room if available	Pass
TC05	CSE-D	60	10-11 AM	No Suitable Classroom Available	Pass
TC06	ECE-B	30	10-11 AM	Room101 Allocated	Pass
TC07	CSE-E	40	9-10 AM	Room102 Allocated only if available	Pass

Test Case ID	Class Name	Students	Time Slot	Expected Output	Status
TC08	ECE-C	20	11-12 AM	Available classroom allocated	Pass

11.Output Screenshots

```
SMART CLASSROOM TIMETABLE ALLOCATION
-----------------------------------
Enter number of classes: 3

Class 1
Enter class name: CSE-A
Enter number of students: 45
Enter subject: Python
Enter time slot (e.g., 9-10 AM):  9-10AM

Classroom Allocated Successfully!
Class: CSE-A
Subject: Python
Classroom: Room101
Time Slot:  9-10AM
Capacity: 50

Class 2
Enter class name: CSE-B
Enter number of students: 35
Enter subject: Java
Enter time slot (e.g., 9-10 AM): 9-10 AM

Classroom Allocated Successfully!
Class: CSE-B
Subject: Java
Classroom: Room101
Time Slot: 9-10 AM
Capacity: 50
```

```
Class 3
Enter class name: ECE-A
Enter number of students: 25
Enter subject: Maths
Enter time slot (e.g., 9-10 AM): 9-10 AM

Classroom Allocated Successfully!
Class: ECE-A
Subject: Maths
Classroom: Room102
Time Slot: 9-10 AM
Capacity: 40

========== FINAL TIMETABLE ==========
Class        Subject      Classroom    Time Slot
-----------------------------------------------------
CSE-A        Python       Room101       9-10AM
CSE-B        Java         Room101      9-10 AM
ECE-A        Maths        Room102      9-10 AM

Program Completed.
```

12.Conclusion

The Smart Classroom Timetable Allocation system provides an efficient way to allocate classrooms based on student strength, classroom capacity, and time-slot availability. It reduces manual work, avoids classroom scheduling conflicts, and improves the utilization of available classrooms. The system automatically assigns suitable classrooms and displays a message when no appropriate classroom is available. Thus, this project provides a simple and reliable solution for classroom timetable management in educational institutions.